

Lacunary gaps answer Shkarin’s positive-density-subset question

Candidate full-solution packet for arXiv:1710.07901

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Status: candidate full positive answer, likely valid. The frequently hypercyclic vector already constructed in Grivaux’s proof has a half-space return set contained in dyadic blocks with infinitely many factor-four gaps. Every subset having a natural density must consequently have density zero. In particular, $T = 2B$ on $\ell^2(\mathbb{N}_0)$ gives an explicit operator answering Question 3.3.

1 Source question

Sophie Grivaux, *Frequently hypercyclic operators with irregularly visiting orbits*, arXiv:1710.07901; J. Math. Anal. Appl. 462 (2018), 542–553 [1]. On page 11 of the source PDF, Question 3.3 asks:

Does there exist an operator $T \in \mathcal{B}(X)$ admitting a frequently hypercyclic vector $x \in X$ such that, for some non-empty open subset $U \subseteq X$, $\mathcal{N}_T(x, U)$ contains no subset A of $\mathbb{N}$ admitting a positive density?

Here

$$\mathcal{N}_T(x, U) = \{n \geq 1 : T^n x \in U\}, \quad \text{dens } A = \lim_{N \rightarrow \infty} \frac{\#(A \cap [1, N])}{N}$$

when the latter limit exists. Thus “positive density” in the question means positive natural asymptotic density, not merely positive upper density.

X which is stable by taking finite unions, such vectors x are easily seen to be such that $\text{dens } \mathcal{N}_T(x, U) = m(U)$ for every non-empty open set $U \subseteq X$ such that $m(\partial U) = 0$, where ∂U denotes the boundary $\overline{U} \setminus U$ of U . Whether this property can hold for *all* non-empty open sets U does not seem clear at all.

Lastly, we recall a question from [20], already mentioned in the introduction:

Question 3.3. — Does there exist an operator $T \in \mathcal{B}(X)$ admitting a frequently hypercyclic vector $x \in X$ such that, for some non-empty open subset $U \subseteq X$, $\mathcal{N}_T(x, U)$ contains no subset A of $\mathbb{N}$ admitting a positive density?

Figure 1: The complete source statement, page 11, Question 3.3.

2 Proof Intuition

Grivaux distributes the centers used to build her frequently hypercyclic vector only in dyadic blocks $[2^j, 2^{j+1})$ whose indices satisfy $j \equiv 0$ or $2 \pmod{5}$. The functional used to detect returns can see only a short interval around each center, and the construction places every center far enough inside its dyadic subblock that this entire interval stays in the same dyadic block. Hence the detected return set is completely empty on the two consecutive blocks with indices $5q + 3$ and $5q + 4$.

For a subset A of this return set, its count is therefore unchanged while the observation scale grows from approximately 2^{5q+3} to 2^{5q+5} , a factor of four. If A had a density δ , the two endpoint subsequences would force both δ and $\delta/4$ as the same limit. Thus $\delta = 0$. This elementary consequence of the source construction is stronger than the “return set has no density” conclusion recorded there.

3 The strengthened theorem

We use exactly the hypotheses of Theorem 1.2 in the published source. A vector x_0 is *bicyclic* for T if it admits a bilateral orbit $(T^n x_0)_{n \in \mathbb{Z}}$ whose linear span is dense.

Theorem 1. *Let $T \in \mathcal{B}(X)$ satisfy the following conditions from Grivaux's Theorem 1.2:*

1. *T has a bicyclic vector x_0 ;*
2. *the associated bilateral-orbit series is unconditionally convergent;*
3. *there are $0 \neq x_0^* \in X^*$ and a finite set $F \subseteq [-d, d] \cap \mathbb{Z}$ such that $\langle x_0^*, T^n x_0 \rangle = 0$ for $n \notin F$.*

Then T admits a frequently hypercyclic vector x such that, for the non-empty open half-space

$$U_0 = \{y \in X : \operatorname{Re} \langle x_0^*, y \rangle > 0\},$$

the return set $\mathcal{N}_T(x, U_0)$ contains no subset of positive natural density. In fact, the same x has this property for every non-empty open $V \subseteq U_0$, and every such $\mathcal{N}_T(x, V)$ has no natural density.

Proof. We audit the particular vector constructed in the proof of Theorem 1.2 of [1]. To keep the return set distinct from the source's center sets, write those center sets as C_s here. The construction starts with dyadic blocks

$$I_j = [2^j, 2^{j+1}) \cap \mathbb{N}$$

and pairwise disjoint subintervals $I_j^{(s)} \subseteq I_j$. Choose $p \geq 1$ so that

$$2^{s+1+p} \geq 2^{s+1} + 2d + 1 \quad (s \geq 1). \quad (1)$$

Inside $I_j^{(s)}$, the set $L_j^{(s)}$ consists of selected integer centers k satisfying

$$\operatorname{dist}(k, \mathbb{N} \setminus I_j^{(s)}) \geq 2^{s+1+p}. \quad (2)$$

The source then defines

$$C_s = \coprod_{\substack{j \geq 2s+p+2 \\ j \in J}} L_j^{(s)}, \quad J = 5\mathbb{N} \cup (5\mathbb{N} + 2). \quad (3)$$

The vector built from these C_s is frequently hypercyclic by the source's Lemma 2.1. Its Lemma 2.2 gives the following exact decomposition of the half-space return set $R := \mathcal{N}_T(x, U_0)$:

$$R = \coprod_{s \geq 1} \coprod_{k \in C_s} I_{k,s}^+, \quad I_{k,s}^+ \subseteq [k - (2^s + d), k + (2^s + d)] \cap \mathbb{N}. \quad (4)$$

If $k \in L_j^{(s)}$, then (1)–(2) imply

$$2^s + d < 2^{s+1+p} \implies I_{k,s}^+ \subseteq I_j^{(s)} \subseteq I_j.$$

Together with (3) and (4), this proves the key containment

$$R \subseteq \bigcup_{j \in J} I_j. \quad (5)$$

For every sufficiently large integer q , neither $5q + 3$ nor $5q + 4$ belongs to J . Consequently (5) makes R empty on

$$[2^{5q+3}, 2^{5q+5}) \cap \mathbb{N}.$$

Let $A \subseteq R$, and put

$$a(N) = \#(A \cap [1, N]), \quad N_q = 2^{5q+3} - 1, \quad M_q = 2^{5q+5} - 1.$$

The gap gives the exact equality $a(M_q) = a(N_q)$. If A has a natural density δ , then

$$\delta = \lim_{q \rightarrow \infty} \frac{a(M_q)}{M_q} = \lim_{q \rightarrow \infty} \frac{a(N_q)}{N_q} \frac{N_q}{M_q} = \frac{\delta}{4},$$

because $N_q/M_q \rightarrow 1/4$. Hence $\delta = 0$. No subset of R can have positive density, proving the first assertion.

Finally let $\emptyset \neq V \subseteq U_0$ be open. Then $\mathcal{N}_T(x, V) \subseteq R$, so it too contains no positive-density subset. Since x is frequently hypercyclic, $\underline{\operatorname{dens}} \mathcal{N}_T(x, V) > 0$. Were its natural density to exist, it would therefore be positive, a contradiction. Thus $\mathcal{N}_T(x, V)$ has no natural density as claimed. $\square$

4 An explicit operator

Corollary 2. *Question 3.3 has an affirmative answer with $X = \ell^2(\mathbb{N}_0)$ and $T = 2B$, where B is the unilateral backward shift.*

Proof. Let $x_0 = e_0$ and define its bilateral orbit by

$$x_{-m} = 2^{-m}e_m \quad (m \geq 1), \quad x_n = 0 \quad (n \geq 1).$$

Then $Tx_n = x_{n+1}$ for every $n \in \mathbb{Z}$, and the negative-index vectors span all coordinate vectors. The bilateral-orbit series is unconditionally convergent (indeed, absolutely in norm), because $\sum_{m \geq 0} \|2^{-m}e_m\| < \infty$. With $x_0^*(y) = y_0$, one has $\langle x_0^*, x_n \rangle = 0$ for every $n \neq 0$. Thus all hypotheses of Theorem 1 hold with $F = \{0\}$. $\square$

The same verification works for λB with $|\lambda| > 1$ on $\ell^p(\mathbb{N}_0)$, $1 \leq p < \infty$, or on c_0 , after replacing 2^{-m} by λ^{-m} . The single explicit Hilbert-space example in Corollary 2 already answers the source question in full.

5 Verification, limitations, and novelty status

The proof uses no numerical approximation. The accompanying exact checker confirms the integer endpoint convention and the convergence of the endpoint ratio to $1/4$ on sample scales; it is a sanity check, not a substitute for the symbolic proof. The decisive source audit is pages 7–8: Lemma 2.2 gives (4), and page 8 gives the margin condition and $J = 5\mathbb{N} \cup (5\mathbb{N} + 2)$. The explicit $2B$ orbit was also checked coordinate by coordinate. The separate verification report records these checks.

A bounded novelty search through 11 August 2026 covered the run registry and attempt indexes and web/arXiv searches for the exact question wording, arXiv identifier, title, “positive density subset”, “irregularly visiting orbit”, and close variants. It found the source and its journal version, but no later paper explicitly resolving Question 3.3. The proof should therefore be treated as a candidate new observation, with moderate novelty confidence pending a specialist citation search.

The theorem does not answer whether *every* frequently hypercyclic operator has such a vector, nor whether one vector has irregular returns to every bounded open set or every open ball. It does give the natural class of all non-empty open subsets contained in one detecting half-space. Human review is recommended, especially of the one-line support containment (5) and the interpretation of “density” as the natural density defined in the source.

References

- [1] S. Grivaux, *Frequently hypercyclic operators with irregularly visiting orbits*, J. Math. Anal. Appl. 462 (2018), 542–553, doi:10.1016/j.jmaa.2018.02.020; arXiv:1710.07901.